

**HIS MAJESTY'S GOVERNMENT
OF NEPAL**

**Ministry of Land Reform
and Management**

**THE REPUBLIC OF
FINLAND**

**Ministry for Foreign Affairs,
Department of International
Development Cooperation**

**WESTERN NEPAL TOPOGRAPHIC
MAPPING PROJECT
Phase I**

**FINAL REPORT
OF
GROUND CONTROL SURVEYS**

VOLUME III: DEFINITION OF GRAVIMETRIC GEOID

**HMG Survey Department of Nepal
FM-International FINNMAP**

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ANNEX I

"GEOID COMPUTATION FOR NEPAL" BY DR MARTIN VERMEER

Geoid computation for Nepal

Martin Vermeer

June 19, 1997

Abstract

A geoid NEPAL97 for the whole of Nepal was computed using material (GPS, levelling, gravimetry) collected over many years in both East and West Nepal. The technique used was "GPS-gravimetric": First a gravimetric geoid was computed using geopotential theory (the MOLODEN-SKII approach). Then, this geoid was "fitted" to a set of given geoid undulation values obtained from GPS heighting and (classical as well as trigonometric) levelling.

The reference surface obtained is a *true* geoid, i.e. a representation of an equipotential surface near sea level, inside the topographic rocks. This means that it can be used together with orthometric heights, as have been traditionally used in Nepal.

The datum of the computed geoid is the *Nagarkot* datum, i.e. in Nagarkot (GPS40) the geoid height is assumed 0.0 m and its height above the reference ellipsoid (GRS80) as determined by GPS is assumed equal to its orthometric height, 2151.78 m. Therefore GPS measurements should be transformed to this datum by adding a constant to all height values h before it is attempted to compute orthometric heights from them using NEPAL97. For the GPS measurements in Eastern Nepal, such a translation had already been made; for the Western Nepal measurements, 22.733 m has to be added.

1 Available material

We obtained from FM INTERNATIONAL OY the material, two gravimetric data files and two topographic height files, the latter constituting a crude digital terrain model of Nepal, good enough for predicting free air anomalies for the nodes of a grid from the rather sparse set of values given. It can be estimated that using terrain information in this way – i.e., first computing point Bouguer anomalies, predicting those to grid nodes, and then converting them back to free air anomalies using external terrain information – will improve the precision of the resulting mean gravity information, and thus of the geoid solution, easily by a factor of two, and at the same time prevent systematic effects due to gravity data point selection being correlated with topography.

We depict the coverage of the available gravimetry in Fig. 1.

2 Data preparation

2.1 The global model EGM96

We evaluated the global model, described in [egm96], for the Nepal area at three elevation levels ("sandwich"): 0 m, 2500 m and 5000 m. Our grid sandwich allowed interpolation in three dimensions of height anomalies as well as gravity accelerations.

2.2 Constructing precise orthometric heights for GPS/gravity points

- We constructed a digital terrain model at 5 x 10 minutes resolution (cell size) into the grid file `htgrid3`. First we gridded the height data given using the software `obingrid`; then we used `ffillin2` with decay parameter 0.999 and a cut-off distance of 1 cell to produce an intermediate file. Finally, a run with decay parameter 0.99 and cut-off distance 10 cells produced `htgrid3`,

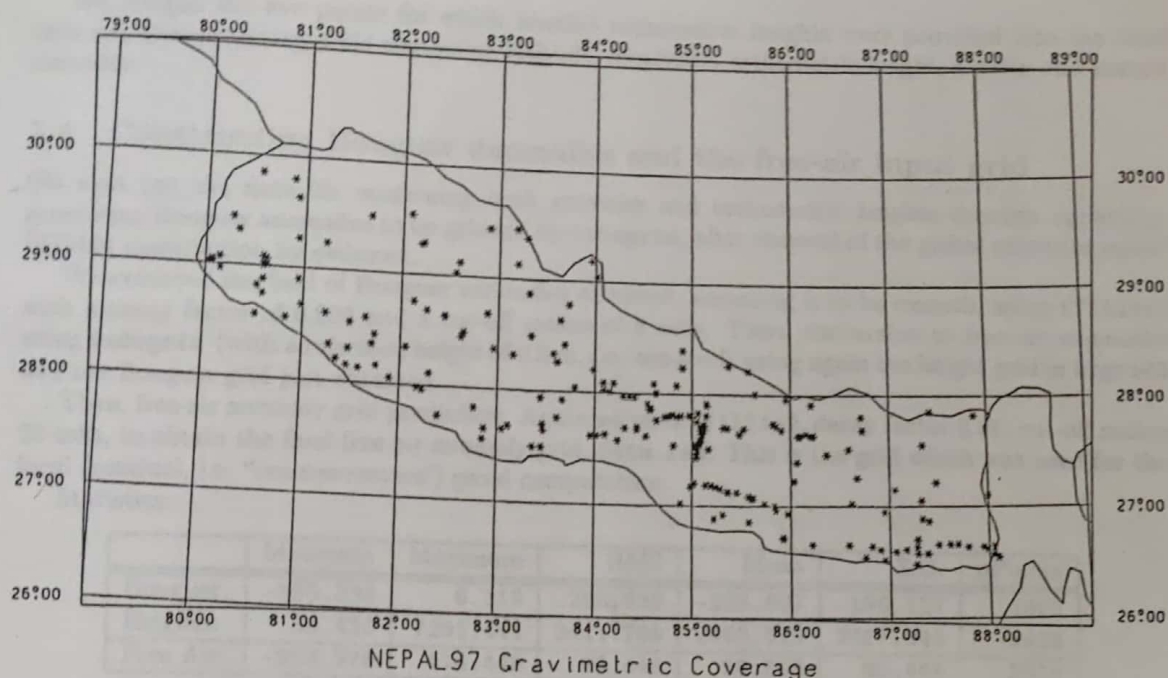


Figure 1: Gravimetric coverage used for the Nepal geoid NEPAL97

used in all subsequent calculations. It constituted a smooth height field covering at least the whole area where also gravimetric data was available.

- For gravity processing, we extracted the records from the West Nepal gravity file `grav_west.dat` having both measured gravity and measured ellipsoidal height into the file `grav_ell_west.dat`.
- From this, we constructed a "stream" file with ellipsoidal heights as data. Using `owinref`, we converted this stream file to one containing orthometric heights, by subtracting N , the geoid undulation, quadratically interpolated from the EGS96 sandwich.
- The point heights obtained turned out to be systematically 18.5 m too big. This is undoubtedly due to the "Ollikainen SPP bias" known to exist for GPS receivers and single point positioning software (M. OLLIKAINEN, personal comm.). This bias is a global property: It has been found in Finland, Denmark and Antarctica also.
There is also a contribution from the fact, that our geoid heights refer to a GRS80 reference ellipsoid with a semi-major axis which is 0.70 m smaller than the originally defined one of 6378137.0 m.
- The orthometric heights ($h + N - 18.5$ m) were put into the file `grav_west_ort_all.dat` manually and used in further work.

2.3 Constructing orthometric heights for gravimetry-only points

We tried to interpolate these heights the height grid file `htgrid3`, constructed as described above. Unfortunately we found, for the five points for which an orthometric height was already given, that the heights thus produced could easily be off by as much as 1000 m – and anyway, the number of heights that could be predicted *at all* in this way was limited to five as well. It is just not worth the trouble.

We merged the five points for which levelled orthometric heights were provided into the total data set, thus obtaining a file containing latitude, longitude, orthometric height, gravity and station identifier.

2.4 Constructing Bouguer anomalies and the free-air input grid

We next ran the data file containing both gravities and orthometric heights through `ogravity`, generating Bouguer anomalies to be gridded by `obingrid`, after removal of the global reference model EGM96 contribution by `owinref`.

We extended the field of Bouguer anomalies obtained, assuming it to be smooth, using `ffillin2` with a decay factor of 0.999 and a cut-off radius of 5 cells. Then, conversion to *free-air anomalies* using `faddgrds` (with a reference height of 0.0 m, i.e. sea level) using again the height grid in `htgrid3` and the Bouguer grid just obtained.

Then, free-air anomaly grid prediction. Again we used `ffillin2`, decay factor 0.95, cut-off radius 20 cells, to obtain the final free air anomaly grid, `data.fa2`. This is the grid which was used for the local (residual, i.e. "remove-restore") geoid computation.

Statistics:

	Minimum	Maximum	RMS	Mean	S.D.	nPoints
Bouguer:	-585.334	6.319	298.939	-225.607	196.127	1805
Heights:	50.430	7291.441	3411.769	2465.819	2357.945	3428
Free Air:	-253.976	414.449	85.446	28.808	80.444	3915

3 Geoid computation

3.1 Generating the mass point solution

We generated a job file by running `fgensol`, which solved from the free air anomalies constructed to obtain a frequency-domain grid file `fsol.zq`. At the same time, a right hand side vector file `fpd9.z` ("predicted data, nine components") was generated for checking. Gravity anomalies (`fdata.pred`) were synthesised from this file; by comparing these with the original data grid contained in `fdata.fa1`, it can be checked that

1. the bias produced by regularization is not too large; and
2. regularization is sufficient to remove any possible instabilities which can be seen as horizontal or vertical "ripples" in the solution.

In this way it is possible to find an acceptable value for each of the regularization parameters. We found the values 10^{30} mGal, i.e., ∞ , for the density constraint, and 1000 E for the horizontal density gradient constraint.

Then the "observable" parameter of the `mgm` system was changed to geoid heights using the `synth` software module, and `fgensol` was run again, this time predicting observables from the solution into the file `fsyn.z`. This file thus contains the *local contribution* to the geoid.

The statistics of the local part of the geoid is summarized in the table in the next sub-section.

3.2 Adding together the global and local parts of the geoid

First the local geoid grid solution was converted to a stream file by running `frfbin`; then this file was run through `owinref` to add the global part. Then the resulted stream file was gridded again with `obingrid`; All this on sea level (height 0.0 m), where the geoid is formally defined.

Finally, changing the free-air geoid (file `geoid.tot`) to the true geoid (file `geoid.true`).

Statistics:

	Minimum	Maximum	RMS	Mean	S.D.	nPoints
Local geoid	-8.229	19.987	5.783	0.794	5.728	4096
Total geoid	-73.040	-0.329	48.435	-43.651	20.989	4096
"True" geoid	-73.043	-3.247	-48.928	-44.763	19.752	4096

One should note that this table was generated for the "central quadrant" of the area of computation; the latter ranged from 22°40' to 33°15' in latitude North and from 74°15' to 95°25' in longitude East. The central quadrant of this area contained the whole country. The area of computation included borders of 50%, to eliminate edge and circularity effects inherent in the FFT method.

3.3 Fit to the GPS/levelling points

We prepared a file *fit.dat* containing the geoid heights as implied by the GPS and levelled heights *h* and *H*.

We ran this through *owinref* to remove the geoid model, and the residuals thus obtained were fitted with *osurf*.

Like in the previous study for Eastern Nepal only, a *quadratic* surface of fit was used, and the origin of the polynomial expansion was *Nagarkot*, GPS40. The Western and Eastern parts of Nepal were treated as independent blocks (i.e. two different constant offsets were assumed in the fit adjustment).

Results:

Latitude	Longitude	Surf.level	Pred.	Error	Resid.	Weighted	Blk
27.1033	88.0006	49.7920	51.1753	0.6123	-1.3833	-1.75	1
27.0200	86.6167	51.2850	50.2008	0.3976	1.0842	1.18	1
28.0836	84.6031	46.5160	44.4322	0.4691	2.0838	2.36	1
27.4650	86.3661	47.4000	47.7057	0.3096	-0.3057	-0.32	1
28.0036	84.6367	45.2080	44.9506	0.4334	0.2574	0.29	1
27.8983	85.1475	43.8960	45.1618	0.4150	-1.2658	-1.39	1
27.6925	85.5236	47.6700	46.3221	0.3392	1.3479	1.43	1
26.8658	87.3947	54.0580	51.9691	0.4878	2.0889	2.39	1
27.4347	85.5972	46.7570	47.7094	0.3304	-0.9524	-1.01	1
27.3722	86.9225	48.8250	48.4681	0.3881	0.3569	0.39	1
27.4953	83.9000	45.1780	46.8214	0.4328	-1.6434	-1.82	1
27.4411	84.8122	46.4280	47.3547	0.3879	-0.9267	-1.01	1
26.7103	85.9269	50.1000	49.2483	0.6103	0.8517	1.08	1
26.8594	85.5697	47.0610	48.6256	0.5215	-1.5646	-1.83	1
26.9303	88.0964	52.5850	52.6140	0.5643	-0.0290	-0.04	1
28.9632	80.1496	26.1270	27.0690	0.3328	-0.9420	-1.00	2
29.2630	80.9378	20.3370	22.9140	0.3636	-2.5770	-2.77	2
28.8222	80.5224	25.1170	26.3302	0.2803	-1.2132	-1.26	2
28.7532	80.5838	25.0360	26.2988	0.2731	-1.2628	-1.31	2
28.6774	80.3316	26.8260	26.9368	0.3369	-0.1108	-0.12	2
28.4290	81.0975	26.2920	25.5421	0.2446	0.7499	0.77	2
28.3301	81.2702	25.1910	25.2831	0.2514	-0.0921	-0.10	2
28.2058	81.3461	25.2570	25.1039	0.3256	0.1531	0.16	2
28.1528	81.4549	24.7150	24.9376	0.3384	-0.2226	-0.24	2
28.3053	81.4676	25.2870	25.0220	0.2214	0.2650	0.27	2
28.5857	81.6371	25.4490	24.2791	0.2204	1.1699	1.20	2
28.6731	80.9808	24.2720	25.5098	0.2347	-1.2378	-1.27	2
29.0646	80.1419	27.5850	26.8247	0.3352	0.7603	0.81	2
29.0540	80.3187	26.3400	26.2863	0.3104	0.0537	0.06	2
29.4842	80.4216	23.6620	23.4594	0.4750	0.2026	0.23	2
29.4748	80.7641	22.4990	21.9719	0.5676	0.5271	0.64	2
29.1917	80.7096	25.7090	24.2687	0.2954	1.4403	1.51	2
28.9918	81.3367	23.6620	23.2511	0.2494	0.4109	0.42	2

28.6110	81.2307	24.7140	25.0869	0.2209	-0.3729	-0.38	2
28.5405	81.4771	25.3240	24.7331	0.2137	0.5909	0.60	2
29.0156	80.6291	25.0490	25.4310	0.2741	-0.3820	-0.40	2
28.3395	82.0546	25.3920	24.1559	0.2144	1.2361	1.27	2
28.0557	81.8946	24.8130	24.4970	0.2793	0.3160	0.33	2
28.3372	81.6158	25.4000	24.8066	0.1999	0.5934	0.61	2
27.9456	81.9223	25.3190	24.3282	0.3919	0.9908	1.08	2
27.8558	82.2534	24.2320	24.0882	0.3700	0.1438	0.15	2
27.7551	82.8053	23.4730	23.9199	0.3105	-0.4469	-0.47	2
27.8715	82.5256	24.4190	24.0538	0.2745	0.3652	0.38	2
28.0624	82.5750	24.6970	23.9928	0.2216	0.7042	0.72	2
28.1118	82.2945	25.0550	24.1900	0.2067	0.8650	0.88	2
28.2479	83.3903	24.9710	22.4309	0.2791	2.5401	2.65	2
27.9216	83.6604	22.6560	23.4823	0.3062	-0.8263	-0.87	2
28.0363	83.7907	22.6810	22.9412	0.3034	-0.2602	-0.27	2
28.1996	83.9839	20.9980	21.7924	0.3243	-0.7944	-0.84	2
28.0224	84.0913	21.8420	22.6969	0.3131	-0.8549	-0.90	2
28.0837	84.1872	21.6860	22.2393	0.3210	-0.5533	-0.58	2
27.9960	84.2574	21.1380	22.6679	0.3212	-1.5299	-1.62	2
27.7673	84.1819	22.6050	23.7087	0.3443	-1.1037	-1.18	2
28.0837	84.6031	22.0800	21.6987	0.3911	0.3813	0.41	2
29.3310	80.6055	24.1080	23.7845	0.3654	0.3235	0.35	2

Residuals Sum of Sqr = 58.3797 SqRt(RSS/(n - m)) = 1.1028

Standard error of unit weight = 1.1028

Blk: 1 2
nPts: 15 40
RMS: 1.2421 0.9386

Solution vector, standard deviations:

Constant <1> (m): 46.3187 +/- 0.3742
Constant <2> (m): 23.5860 +/- 0.4617

Diff.<1>-<2>(m): 22.7328 +/- 0.6217

Lat tilt (m/deg): -6.4642 +/- 1.3483
Lon tilt (m/deg): -0.3999 +/- 0.4344
Torsion (m/deg2): -2.9553 +/- 1.0070
Lat curv(m/deg2): -4.5703 +/- 1.5299
Lon curv(m/deg2): -0.1127 +/- 0.1704

oSurFit: (Obs=55/Unkn=7/Fix=0/Delta= 0.0) End.

We can see several things from this table.

- We performed a separate fit for the GPS campaign in Eastern Nepal (block 1) and that in Western Nepal (block 2). This was necessary to account for the difference in datum between the results of the two campaigns. Whereas the Eastern results were given in the Nagarkot datum (i.e. making the ellipsoidal height of this point equal to its levelled orthometric height of 2151.78 m), the Western Nepal results were in a free floating system differing by some 22 m from this. The precise difference was estimated in the simultaneous fit to be 22.73 ± 0.62 m.
- The residuals are all below 3 m and the weighted residuals below 3, i.e., residuals are under 3σ . Especially the Western Nepal GPS and levelling results appear to be of very good quality,

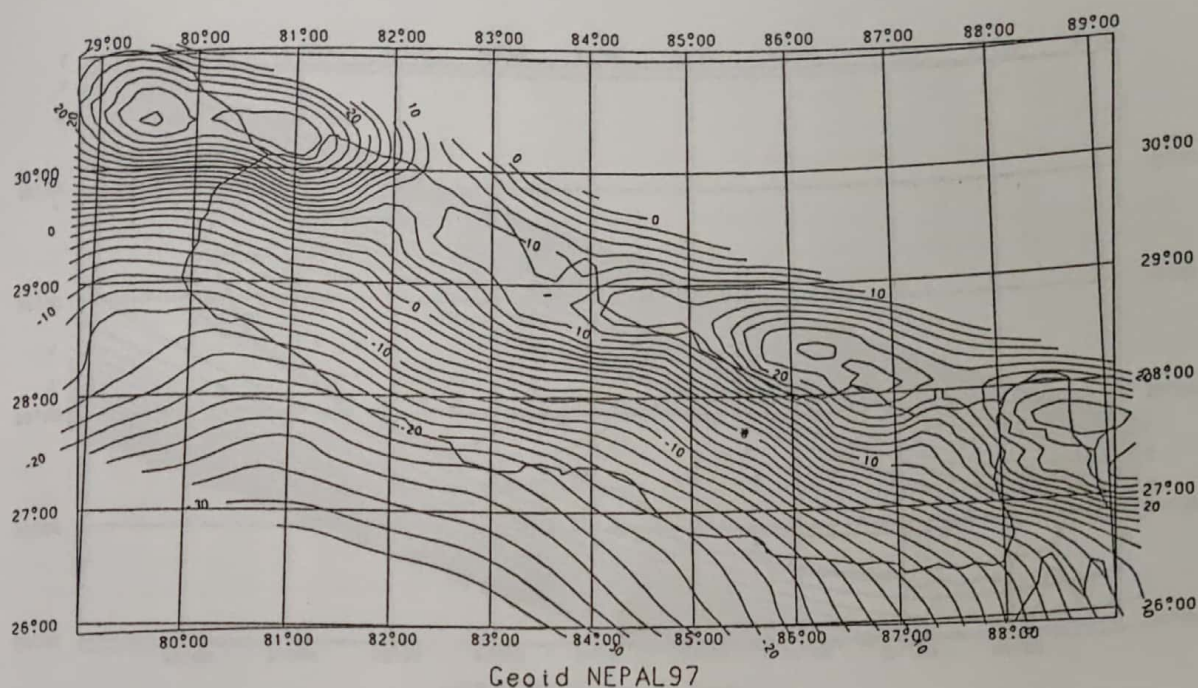


Figure 2: The geoid of Nepal NEPAL97. Nagarkot datum (datum point indicated). Unit m. Contour interval 2 m.

slightly better than the Eastern Nepal results. This may be due to the Eastern part being more mountainous.

- The overall S.D. of unit weight is ± 1.10 m, slightly poorer than the ± 1.05 m found earlier for Eastern Nepal only. This is undoubtedly due to the additional constraint of processing the two areas together. Individually the RMS of Western Nepal is very good, ± 0.94 m, clearly better than Eastern Nepal, ± 1.24 m.

3.4 Bringing the geoid into the Nagarkot datum

We finally applied a straight vertical translation to the geoid map obtained, in order to make the geoid undulation for the location Nagarkot (at latitude $27^{\circ}41'33''$, longitude $85^{\circ}31'25''$) vanish. This translation amounted to $+1.368$ m. Also the graphic representation (Fig. 2) was plotted after this datum transformation, from the final geoid grid geoid.40.

3.5 Geoid results in graphic form

In Fig. 2 we display the geoid map obtained. We note that the digital coastline and border map used for plotting was very crude and contained clear errors.

The formal error obtained from the fit is displayed in Fig. 3.

3.6 The accuracy of the gravimetric geoid

From the GPS/levelling fit, it appears that the quality of the gravimetric geoid is around ± 1 m. We will try to establish an error budget for this value. Firstly, it is clear that many of the heights in the very mountainous terrain of Nepal may be of this accuracy level, especially if they were determined by trigonometry.

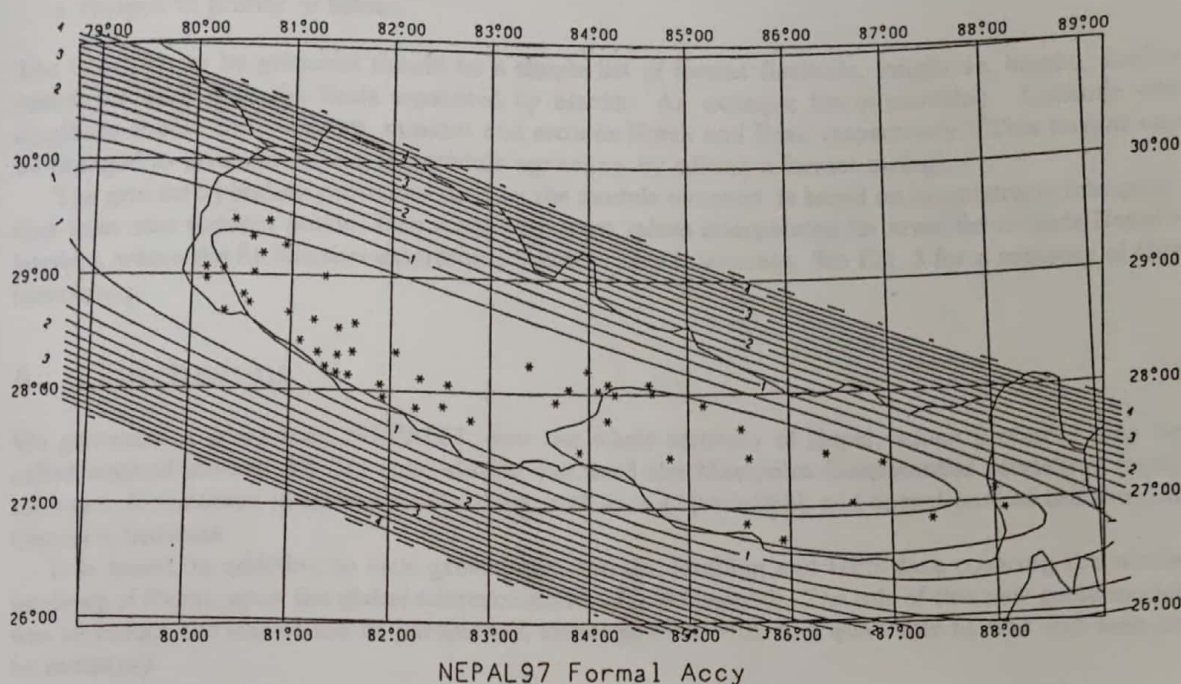


Figure 3: Geoid of Nepal NEPAL97 formal accuracy as derived from GPS/levelling fit (support points indicated). Unit m. Contour interval 0.25 m.

But also the gravimetric geoid is inaccurate. VERMEER [verm95a] gives a formula for the theoretical precision of a gravimetric geoid:

$$\sigma_N [\text{mm}] = 0.4d [\text{km}] \sigma_{\Delta g} [\text{mGal}] \sqrt{\ln \frac{D}{d}},$$

where d is the spacing of gravimetric data points, D that of GPS/levelling points and $\sigma_{\Delta g}$ the "error of prediction" of gravity anomaly values over a distance d .

From this one can deduce, for the plausible values (for Nepal) $d = D = 50$ km and $\sigma_{\Delta g} = 50$ mGal, that $\sigma_N = 1$ m.

4 The grid interpolator

Besides the graphical representations presented here, also a digital grid with accompanying interpolation software was prepared. The software runs under MS-DOS and is called simply NEPAL97. The package – consisting of a large number of small batch files calling each other, and three Turbo Pascal program modules doing the actual work, and a large number of small configuration text files – offers the options of:

- Editing the input file points.txt;
- Inspecting or editing the output file points.out;
- Computing geoid heights for the coordinate positions given in the input file;
- Adding or subtracting geoid undulations to/from the heights already present in the input file; In this case, the height type in the file points.txt should be correct, i.e., orthometric when adding, ellipsoidal when subtracting. *The software does not check.*

- Output to printer or screen.

The input file to be prepared should be a simple list of format (latitude, longitude, height, station name) for each line, the fields separated by blanks. An example file is provided. Latitude and longitude should be in degrees, minutes and seconds North and East, respectively. (This format can be changed in principle within the module ogravity by editing a format string).

The grid interpolation procedure used by the module owinref is based on bi-quadratic interpolation from nine support points. One should not trust values interpolated for areas far outside Nepal's borders, where the fit function determined becomes very inaccurate. See Fig. 3 for a measure of this inaccuracy.

5 Conclusions

We generated a geoid map, NEPAL97, over the whole territory of Nepal, which probably may be called state-of-the-art. For its computation was used the Masspoint Geopotential Modelling (mgm) software, documented in several publications, such as, e.g. [verm95b], and operational at the Finnish Geodetic Institute.

It is based, in addition to local gravimetric, terrain, levelling and GPS data covering the whole territory of Nepal, upon the global reference model EGM96 [egm96]. The role of this new geoid model was to reduce the size of the fit parameters, although even with it, a quadratic fit was still seen to be necessary.

Acknowledgements

The EGM96 model has been computed by a US consortium and made available to the world free of charge. In order to make future models even better, it would be desirable to provide the consortium with the kind of feedback as is contained in the present report. The author leaves this as a suggestion to its intended audience.

This report was authored in LyX, the graphic L^AT_EX 2_ε screen editor for Linux.

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LIST OF OBSERVATIONS OF WESTERN NEPAL

WESTERN NEPAL TOPOGRAPHIC MAPPING PROJECT
LIST OF GRAVITY OBSERVATIONS

GPS-POINTS

No	B (dms)			L (dms)			Ellips. h(m)	Orthom. h(m)	Gravity (mgal)
1	80	4	41.8698	28	51	3.7948	150.480		
2	80	8	58.6658	28	57	47.6306	180.469	216.465	979008.24
3	80	56	16.2090	29	15	46.7167	549.984	577.146	978929.06
4	80	31	20.7143	28	49	19.8816	162.040	199.002	978992.82
5	80	35	1.5735	28	45	11.3387	149.689	187.525	978990.55
6	80	31	6.2135	28	33	57.2388	121.245		979000.24
7	80	19	53.6651	28	40	38.7659	130.702	169.657	
8	80	14	47.1054	28	47	42.8762	140.154		
9	80	49	33.9190	28	33	8.0130	114.324		978990.82
10	81	5	51.1575	28	25	44.5059	104.955	144.067	
11	81	16	12.7835	28	19	48.4097	119.099	159.873	978968.15
12	81	20	46.0835	28	12	20.8309	110.590	152.453	978971.72
13	81	27	17.5864	28	9	10.1335	100.477	142.910	978969.84
14	81	28	3.2877	28	18	19.2579	131.595	171.464	978966.83
15	81	38	13.4719	28	35	8.4342	645.703	678.247	978895.36
16	80	58	50.7742	28	40	23.0097	141.213	178.533	978980.71
17	80	8	31.0082	29	3	52.5813	224.885	257.238	979008.35
18	80	19	7.4948	29	3	14.3517	1376.049	1408.745	
19	80	19	39.9231	29	18	55.5416	766.689		978918.40
20	80	15	54.8783	29	25	57.8755	1240.397		978823.59
21	80	25	17.8218	29	29	3.0717	2510.983	2534.059	
22	80	45	50.6929	29	28	29.4098	2829.243	2850.072	
23	80	56	44.9571	28	53	46.8868	1276.029		978786.51
24	81	0	27.1497	29	2	20.2897	1390.118		
26	80	42	34.6662	29	11	30.0632	2658.818	2684.712	
27	81	13	3.0198	29	14	9.8047	572.685		978913.69
29	81	20	12.0195	28	59	30.4793	2225.122	2252.632	
31	81	13	50.5743	28	36	39.7242	153.456	190.148	978979.14
32	81	28	37.5057	28	32	25.7799	1550.613	1586.124	978686.10
33	81	48	59.8151	28	43	11.4173	2763.370		
34	82	3	45.0587	28	41	54.5136	2151.054		
35	82	15	42.8371	28	40	40.1548	1848.344		978621.50
36	80	37	44.6547	29	0	56.1882	1308.970	1340.296	
37	82	3	16.5838	28	20	22.2411	2375.279	2407.815	978519.80
38	82	23	37.8638	27	44	33.8896	233.622		978933.03
39	82	6	37.6486	28	12	8.7759	628.691		
40	81	53	40.7354	28	3	20.4207	124.594	164.766	978974.12
41	81	36	57.0581	28	20	13.7828	985.550	1023.412	
42	81	43	7.1414	27	59	1.6856	91.875		
43	81	55	21.7049	27	52	3.9920	89.677		
44	81	55	20.1887	27	56	44.0882	408.618	449.559	
45	82	15	12.1208	27	51	20.8835	821.944	863.086	
46	82	31	49.7720	27	41	27.8342	189.874		
47	82	48	19.0316	27	45	18.4803	938.830	981.000	
48	82	47	23.0551	27	50	48.9945	274.904		
49	82	31	32.0865	27	52	17.2393	218.045	257.352	
50	82	34	30.0924	28	3	44.6148	1672.750	1707.492	
53	82	17	40.3638	28	6	42.6542	598.490	634.027	978877.63
54	82	57	23.3715	28	21	35.0990	3604.069		978230.88
55	83	9	8.0730	28	18	52.5794	1714.512		978640.38
56	83	1	2.8471	28	5	11.5079	1985.573		
57	83	8	18.7423	27	55	8.0312	1816.409		
59	83	23	25.0603	28	14	52.4185	3138.994	3163.246	
60	83	26	31.4163	27	56	25.0193	588.155		978862.47
61	83	39	37.4882	27	55	17.6670	515.838	549.683	978958.41
62	83	49	37.7712	27	50	51.5685	1270.105		
63	83	47	26.5786	28	2	10.6957	881.683	912.704	978790.77
64	83	48	7.9843	28	9	51.6621	983.403		
65	83	59	1.9489	28	11	58.6662	791.699	819.083	978785.58

66	84	5	28.5554	28	1	20.7918			
67	84	11	14.0101	28	5	1.4152	472.948	504.637	978859.17
68	84	15	26.5411	27	59	45.4507	1252.000	1281.642	978675.50
69	84	10	54.7526	27	46	2.4078	334.942	367.460	
72	84	27	51.1336	28	24	3.2192	1896.328	1933.445	
73	84	29	39.2693	28	13	35.4483	3921.863		
74	84	36	11.0003	28	5	1.3845	1888.354		
75	85	3	13.6303	28	29	20.1926	1946.338	1972.741	978509.30
76	80	36	19.6502	29	19	51.6956	3005.321		
77	82	40	9.6869	28	16	13.6902	1657.210	1682.450	
78	81	34	35.5252	28	8	2.2838	2058.614		
79	82	37	12.0911	28	27	16.9050	107.944	149.882	978968.09
80	80	23	.8837	29	44	16.0180	2518.758		978495.17
81	80	32	34.5092	29	50	38.5130	1101.547		
82	80	44	1.3482	29	58	41.6137	873.381		978858.75
83	80	58	55.7554	30	8	2.7705	2494.597		
84	80	52	23.1360	29	47	40.2462	3630.285		
85	80	55	30.5069	29	37	41.4591	2448.444		978551.79
86	81	1	30.8878	29	29	31.5775	2553.282		978531.90
87	81	12	58.1124	29	32	26.2556	1871.263		
88	81	17	15.9581	29	45	11.4542	1790.753		
89	81	31	20.1540	30	4	52.6228	2525.527		
91	82	2	40.8466	30	6	41.3862	3175.464		
92	81	49	15.1949	29	58	15.4777	3899.428		
93	81	23	18.8903	29	42	6.5138	2962.110		
94	81	20	57.8304	29	37	5.2227	2799.681		
95	81	40	12.5616	29	30	11.9773	2626.256		
96	81	25	51.8245	29	29	44.1403	1386.292		978722.43
97	81	27	28.0663	29	12	28.7103	2246.929		
98	81	43	13.6673	29	8	28.0762	2450.953		
99	82	2	3.1175	28	53	54.7004	3048.590		
100	82	5	14.1232	29	4	54.6823	2275.293		
101	81	55	26.9638	29	12	57.7195	3027.028		
102	82	11	30.5610	29	16	25.5404	2541.249		
103	82	4	55.5780	29	32	21.0434	2358.166		978544.20
104	81	57	13.9151	29	38	16.4767	2962.007		978340.61
105	82	18	21.7540	29	54	43.3362	2623.730		
106	82	15	35.1233	29	35	5.6412	3531.677		
107	82	27	21.9893	29	34	53.8424	2768.359		
108	82	22	11.8902	29	17	21.2634	2950.780		
109	82	35	27.8920	29	7	16.1115	3117.744		
110	82	33	21.6915	29	1	29.7018	3018.470		978381.87
112	82	27	4.9387	28	38	13.3318	2133.598		978538.60
113	82	55	32.0829	28	31	2.4015	1554.956		978696.74
114	82	53	4.2142	28	40	19.5278	2812.504		
115	82	49	16.3960	28	59	7.8250	2411.999		978500.61
117	82	53	55.9578	29	26	33.9432	2486.538		
118	83	10	54.8243	29	27	.4678	3933.392		978171.10
119	82	59	42.6385	28	52	21.4687	3887.169		978139.27
120	83	18	47.7673	28	52	19.0649	2816.691		
121	83	20	12.8463	28	29	30.1316	3952.453		978171.14
122	83	41	5.5325	28	25	8.9228	2017.506		978557.43
123	83	38	5.9362	28	38	12.4546	2536.848		978450.26
124	83	43	28.6545	28	46	50.6461	2447.276		978445.06
125	83	51	45.6977	29	0	7.5039	2723.618		978374.60
126	84	0	51.5859	29	2	30.1171	3471.239		
127	83	57	42.0171	29	10	56.4845	3923.629		978114.63
128	84	5	28.9787	28	38	28.5083	3795.028		978114.50
129	84	20	28.3241	28	32	8.8801	3370.312		
130	84	28	22.4983	28	38	5.5112	2149.731		
131	84	38	26.7967	28	35	28.8629	3693.243		
133	84	50	2.7087	28	31	28.6458	3547.569		
134	85	6	28.8535	28	36	46.4786	2391.755		
136	83	11	24.3760	29	7	48.7491	3810.685		
137	83	17	15.4331	29	22	14.6535	4084.026		978140.24
138	82	56	25.8877	29	10	21.8541	4159.879		978083.62
				28	20	7.6301	3617.633		
							1756.363		

WESTERN NEPAL TOPOGRAPHIC MAPPING PROJECT
LIST OF GRAVITY OBSERVATIONS

OLD BRITISH AND OTHER POINTS

No	B (dms)	L (dms)	Ellips. h(m)	Orthom. h(m)	Gravity (mgal)
201	84 38 .6	27 45 43.6			
202	85 9 45.5	27 30 37.5		1945.3	978518.50
203	83 36 24.6	27 45 53.1		2505.4	978400.40
204	82 38 32.7	28 29 42.5		1845.4	978579.40
205	82 15 42.8	28 40 40.1		3052.3	978375.80
301	83 36 36.	28 15 42.		1870.8	978621.50
302	83 34 6.	28 20 6.			978762.50
303	83 27 0.	27 30 12.			978772.80
304	83 27 42.	27 42 12.			978901.10
305	82 26 36.	28 37 36.			978906.50
306	82 13 6.	29 17 6.			978871.90
307	82 10 36.	27 59 0.			978543.50
308	84 33 42.	27 51 30.			978960.60
309	82 45 42.	27 49 42.			978870.80
310	83 3 6.	27 38 48.			978930.40
311	82 38 30.	28 29 42.			978919.80
312	82 7 36.	28 45 18.			978375.80
313	81 28 36.	28 32 24.			978701.20
314	82 3 18.	28 20 24.			978701.20
315	82 52 0.	27 38 48.			978519.80
316	83 57 18.	27 36 48.			978920.10
317	81 43 54.	28 26 18.			978892.20
318	81 45 12.	28 9 36.			978908.50
319	81 42 24.	28 21 18.			978964.50
320	82 13 36.	27 57 42.			978952.80
321	80 35 0.	28 52 54.			978959.70
322	80 35 0.	29 0 0.			978976.70
323	80 35 0.	29 5 0.			978815.40
324	80 37 42.	29 0 42.			978734.60
325	80 37 0.	29 4 0.			978800.10
326	80 3 36.	29 0 48.			978781.30
327	80 3 36.	29 1 12.			978798.40
328	80 3 48.	29 1 36.			978692.20
329	80 35 18.	29 1 48.			978695.50
330	80 3 36.	29 2 6.			978701.30
331	80 3 42.	29 2 30.			978754.60
332	83 57 42.	27 36 54.			978639.70
333	84 0 54.	27 37 6.			978892.40
334	84 6 6.	27 37 48.			978889.60
335	83 21 42.	27 30 18.			978883.30
336	82 51 6.	27 33 12.			978907.40
337	83 6 30.	27 40 48.			978919.40
338	83 28 6.	27 39 6.			978917.60
					978903.90

ANNEX K

LIST OF OBSERVATIONS OF EASTERN NEPAL

Eastern Nepal Topographic Mapping Project
List of gravity observations.

GPS-points

Station	B(dms)	L(dms)	Ellipsoidal h(m)	Ortho H(m)	gravity
GPS 30	27 49 15	87 52 02			
GPS 34	27 06 12	88 00 02	4291.69		977 965.14
GPS 21	27 01 12	86 37 00	3644.13	3631.6	978 161.47
GPS125	27 50 59	87 25 01	2343.49	2344.4	978 443.90
GPS 26	27 33 10	87 20 20	2783.24		978 247.37
GPS 28	27 12 47	87 28 16	2025.00		978 466.77
GPS 6	28 05 01	84 36 11	2797.58		978 358.10
GPS 8	28 12 41	85 34 10	1970.73	1972.7	978 506.08
GPS 16	27 27 54	86 21 58	3872.00		978 196.16
GPS 19	27 48 09	86 43 04	3362.84	3356.3	978 235.70
GPS 36	27 36 49	84 01 05	3573.91		978 158.10
GPS 38	27 00 50	84 52 51	145.17		978 889.18
GPS102	28 23 28	84 54 04	73.06		978 869.98
GPS103	28 14 04	84 54 09	1616.87		978 506.10
GPS105	28 04 25	84 50 28	1934.18		978 496.04
GPS107	28 15 28	85 22 11	911.68		978 720.58
GPS108	27 41 27	84 18 28	1735.43		978 477.50
GPS109	28 00 13	85 38 16	161.73		978 888.90
GPS112	28 00 13	84 38 12	2625.34		978 357.60
GPS113	27 54 42	86 12 41	1516.69	1521.9	978 609.18
GPS117	27 30 53	86 35 10	1979.22		978 438.92
GPS118	27 41 09	86 43 44	2478.76		978 433.13
GPS119	27 57 32	85 57 25	2799.93		978 324.42
GPS120	27 15 10	86 40 21	1728.67		978 512.62
GPS130	27 32 22	84 48 29	1232.08		978 685.12
GPS132	27 37 12	84 31 37	286.06		978 866.28
GPS134	27 40 39	84 25 55	188.65		978 875.10
GPS136	27 34 36	84 13 50	192.70		978 880.85
GPS142	27 10 49	85 11 18	135.15		978 888.57
GPS172	27 54 16	84 32 34	146.14		978 854.07
GPS147	26 55 13	85 18 23	312.97		978 869.47
GPS 39	27 53 54	85 08 51	64.72		978 864.36
GPS 40	27 41 33	85 31 25	570.34	572.9	978 810.00
GPS 41	27 44 25	85 49 17	2151.78	2151.8	978 494.14
GPS 42	27 38 07	86 13 58	1383.97		978 636.39
GPS 47	26 29 01	87 16 01	1927.35		978 532.30
GPS 48	26 51 57	87 23 41	67.04		978 878.38
GPS114	27 23 37	86 03 50	2050.54	2047.6	978 492.20
GPS122	27 08 49	87 03 12	493.08		978 830.65
GPS126	27 36 40	85 06 10	1214.56		978 696.97
GPS141	27 09 44	84 59 03	2329.76		978 459.60
GPS144	27 03 50	85 35 39	113.42		978 863.54
GPS160	26 42 14	87 16 46	125.37		978 855.44
GPS 10	27 26 05	85 35 50	129.89		978 856.50
GPS 23	27 22 20	86 55 21	2624.69	2631.7	-
GPS 35	27 29 43	83 54 00	3628.94	3619.6	-
GPS 37	27 26 28	84 48 44	813.26	834.2	-
GPS 45	26 42 37	85 55 37	710.38	725.1	-
GPS 43	26 51 34	85 34 11	55.70	70.1	-
GPS169	26 55 49	88 05 47	64.12	82.3	-
			1872.81	1863.0	-

OTHER GRAVITY STATIONS

Station	B(dms)	L(dms)	Ellipsoidal h(m)	Ortho H(m)	gravity
OTH 1	27 38 42	85 28 42		1348.6	978 661.99
OTH 2	27 38 42	85 28 48		1592.0	978 616.84
OTH 3	27 54 54	85 08 30		526.2	978 813.66
OTH 4	28 00 00	84 37 00		1025.5	978 728.03
OHT 5	27 51 18	84 33 54		264.9	978 870.79
OTH 6	27 42 00	84 26 00		205.5	978 880.91
OTH 7	27 26 00	85 02 30		459.5	978 831.90
OTH 8	27 07 24	85 21 12		131.9	978 854.86
OTH 9	27 02 58	87 19 37		2106.4	978 491.11
OTH 10	27 41 50	85 21 29		1332.1	978 661.30

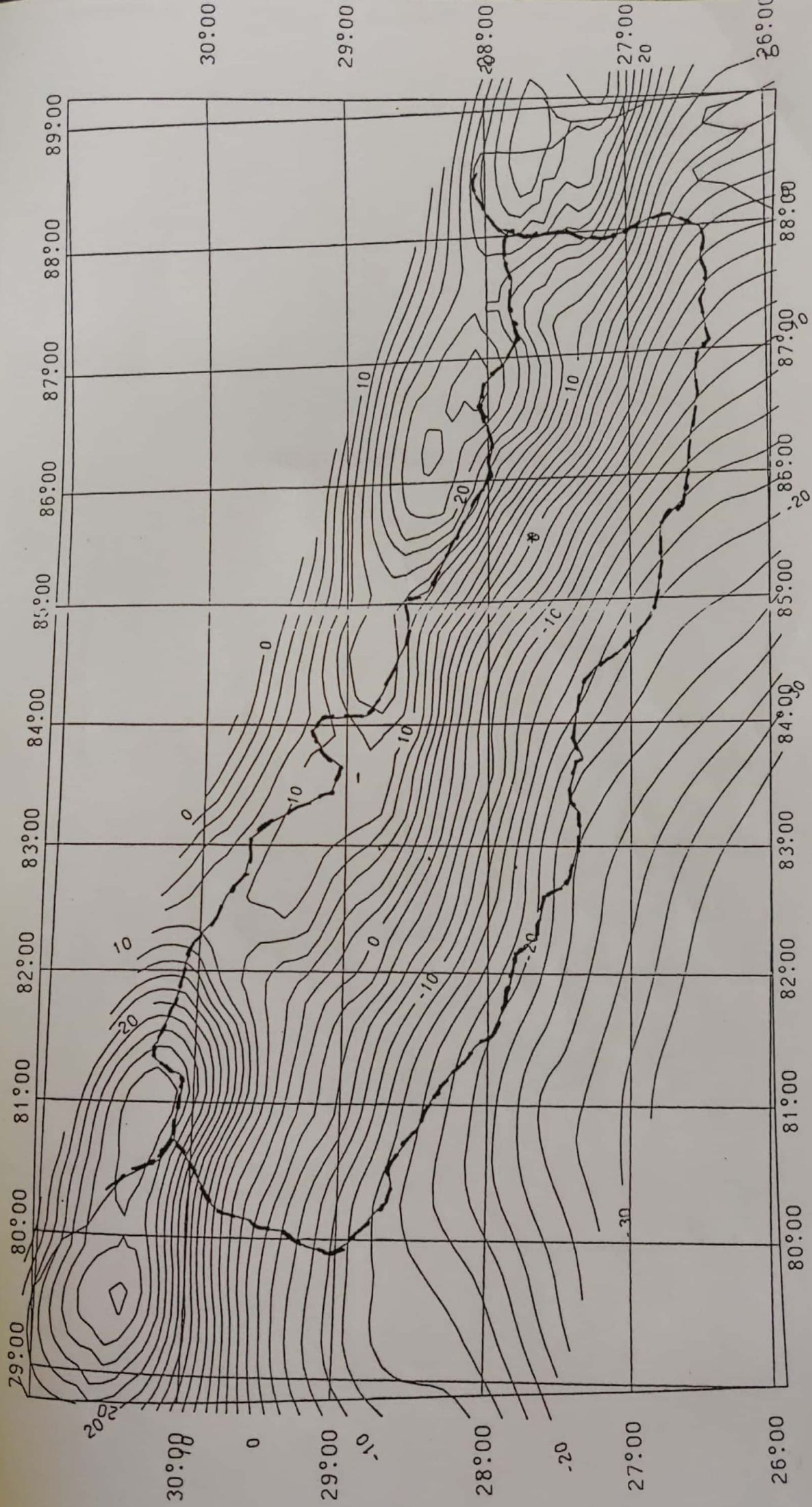
OLD BRITISH POINTS

Station	B(dms)	L(dms)	Ellipsoidal h(m)	Ortho H(m)	gravity
A1 1	27 42 12	85 19 18		1297.6	978 666.54
A1 2	26 32 54	88 05 30		87.0	978 888.85
A1 3	27 00 54	84 52 42		82.0	978 873.05
A1 4	26 42 37	85 55 37		70.1	978 866.36
B2 1	26 57 00	86 56 53		1782.0	978 553.80
B2 2	26 51 57	87 23 41		2047.0	978 492.20
B2 3	27 13 24	86 02 02		1898.0	978 525.90
B2 4	26 53 12	87 20 00		1455.0	978 617.80
B2 5	28 05 01	84 36 11		1973.0	978 509.30
B2 6	27 45 44	84 38 00		1945.0	978 518.50
B2 7	27 48 32	85 08 28		2068.0	978 506.90
B2 8	27 30 36	85 09 48		2500.0	978 400.40
B2 9	27 41 06	85 07 42		745.2	978 775.90
B2 10	27 49 54	85 03 24		568.6	978 807.20
B2 11	27 45 24	85 02 48		501.5	978 823.00
B2 12	27 48 06	84 59 36		336.1	978 853.20
B2 13	27 48 30	84 58 12		327.8	978 855.80
B2 14	27 48 18	84 53 12		329.7	978 859.00
B2 15	27 48 00	84 50 30		291.7	978 865.70
B2 16	27 48 06	84 45 06		242.6	978 872.90
B2 17	27 48 48	84 48 00		277.2	978 862.70
B2 18	27 48 24	84 46 30		267.7	978 867.20
B2 19	27 47 42	84 43 06		234.0	978 870.80
B2 20	27 51 48	84 34 54		180.2	978 866.50
B2 21	28 05 01	84 11 14		1281.0	978 675.50
B3 1	27 43 36	85 06 18		1150.0	978 695.10
B3 2	27 41 12	85 06 00		1568.0	978 614.10
B3 3	27 40 54	85 04 54		1675.0	978 593.10
B3 4	27 39 30	85 06 06		1863.0	978 556.30
B3 5	27 38 42	85 05 42		1712.0	978 583.50
B3 6	27 37 00	85 05 06		1939.0	978 533.70
B3 7	27 36 42	85 05 48		2223.0	978 471.00
B3 8	27 42 24	85 05 24		1461.0	978 634.60
B3 9	27 43 30	85 07 30		969.0	978 731.20
B3 10	27 35 06	85 05 00		2304.0	978 452.70
B3 11	27 34 24	85 04 48		2031.0	978 507.00
B3 12	27 33 30	85 04 12		1824.0	978 551.70
B3 13	27 33 48	85 04 06		1647.0	978 589.20

Station	B(dms)	L(dms)	Ellipsoidal h(m)	Ortho H(m)	gravity
B3 14	27 32 54	85 02 00		1199.0	978 683.70
B3 15	27 32 06	85 02 18		1104.0	978 708.00
B3 16	27 32 00	85 02 48		825.0	978 760.80
B3 17	27 31 00	85 03 12		622.0	978 800.20
B3 18	27 28 24	85 02 54		431.0	978 826.80
B5 1	27 44 18	85 50 00		1327.4	978 650.50
B5 2	27 43 06	85 51 30		1742.4	978 574.40
B5 3	27 41 00	85 54 30		2204.8	978 481.90
B5 4	27 37 30	86 04 30		1240.5	978 662.80
B5 5	27 37 42	86 04 24		1064.9	978 696.00
B5 6	27 37 30	86 05 36		1125.3	978 685.20
B5 7	27 38 24	86 06 24		1646.3	978 586.00
B5 8	27 39 00	86 09 48		2174.2	978 482.30
B5 9	27 38 06	86 11 18		2476.1	978 419.90
B5 10	27 37 36	86 12 54		2150.0	978 480.70
B6 1	27 57 42	84 24 36		326.0	978 859.20
B6 2	27 58 06	84 21 12		567.0	978 818.50
B6 3	27 58 12	84 17 36		256.0	978 881.50
B6 4	28 12 06	83 59 00		409.0	978 855.20
B6 5	28 00 30	84 09 12		404.0	978 859.70
B6 6	28 05 18	84 05 06		446.0	978 849.90
B6 7	26 37 00	87 10 06		90.0	978 870.80
B6 8	26 35 54	87 03 36		82.0	978 872.10
B6 9	26 31 18	86 56 48		76.0	978 875.80
B6 10	26 36 54	86 54 18		88.0	978 876.10
B6 11	26 38 48	86 49 30		117.0	978 874.90
B6 12	26 39 24	86 35 06		120.0	978 868.40
B6 13	26 32 44	86 45 04		82.0	978 881.70
B7 1	26 38 36	87 56 42		119.0	978 858.30
B7 2	26 37 18	87 59 12		114.0	978 861.00
B7 3	26 35 00	88 03 18		96.0	978 876.70
B7 4	26 33 12	88 05 36		86.0	978 886.80
B7 5	26 39 12	88 03 06		127.0	978 859.70
B7 6	26 38 36	87 47 36		130.0	978 847.00
B7 7	26 42 18	87 17 00		150.0	978 856.50
B7 8	26 38 36	87 16 24		100.0	978 862.70
B7 9	26 34 54	87 17 00		87.0	978 869.80
B7 10	26 34 54	87 22 42		138.0	978 850.40
B7 11	26 39 42	87 29 36		144.0	978 846.80
B7 12	26 39 54	87 36 36		121.0	978 849.00
B7 13	26 39 54	87 41 06		127.0	978 848.60
B7 14	27 12 00	85 01 00		150.0	978 859.90
B7 15	27 11 00	85 08 00		156.0	978 856.50
B7 16	27 10 00	85 14 00		147.0	978 854.50
B7 17	27 09 00	85 17 00		147.0	978 855.30
B7 18	27 07 00	85 22 00		128.0	978 855.10
B7 19	26 53 00	85 13 00		79.2	978 865.90
B7 20	27 07 00	85 27 00		134.0	978 858.60
B8 1	27 05 00	85 34 00		142.0	978 853.90
B8 2	27 04 00	85 37 00		157.0	978 852.90
B8 3	26 51 30	85 34 12		81.0	978 861.60
B8 4	27 01 00	85 44 00		184.0	978 846.20
B8 5	27 00 00	85 50 00		208.0	978 843.00
B8 6	26 58 00	85 51 00		206.0	978 844.10
B8 7	26 56 00	85 58 00		159.0	978 847.10
B8 8	26 43 34	85 55 25		75.9	978 864.90
B8 9	26 39 35	86 13 04		72.0	978 872.80

ANNEX L

GEOID CHART "GEOID NEPAL 97"



Geoid NEPAL97

ANNEX M

HEIGHTS OF GPS - POINTS

201/51

WESTERN NEPAL TOPOGRAPHIC MAPPING PROJECT / GPS IN 1996-1997
FINAL COORDINATES OF GPS-POINTS IN NEPALESE SYSTEM

POINT	LONGITUDE			LATITUDE			HEIGHT	
	DEG	MIN	SEC	DEG	MIN	SEC	ELLIPS.	ORTHO
1	80	4	41.8691	28	51	3.7960	150.424	187.191
2	80	8	58.6652	28	57	47.6315	180.414	216.465
3	80	56	16.2094	29	15	46.7160	549.935	577.146
4	80	31	20.7138	28	49	19.8824	161.984	199.002
5	80	35	1.5730	28	45	11.3395	149.633	187.525
6	80	31	6.2128	28	33	57.2401	121.190	160.520
7	80	19	53.6644	28	40	38.7672	130.648	169.657
8	80	14	47.1048	28	47	42.8773	140.100	175.114
9	80	49	33.9185	28	33	8.0141	114.266	152.900
10	81	5	51.1570	28	25	44.5069	104.891	144.067
11	81	16	12.7830	28	19	48.4108	119.035	159.873
12	81	20	46.0829	28	12	20.8320	110.529	152.453
13	81	27	17.5857	28	9	10.1347	100.415	142.910
14	81	28	3.2873	28	18	19.2588	131.541	171.464
15	81	38	13.4719	28	35	8.4343	645.624	678.247
16	80	58	50.7740	28	40	23.0103	141.153	178.533
17	80	8	31.0078	29	3	52.5820	224.830	257.238
18	80	19	7.4945	29	3	14.3522	1375.997	1408.745
19	80	19	39.9233	29	18	55.5415	766.640	794.202
20	80	15	54.8785	29	25	57.8751	1240.347	1265.813
21	80	25	17.8221	29	29	3.0712	2510.936	2534.059
22	80	45	50.6935	29	28	29.4089	2829.196	2850.072
23	80	56	44.9570	28	53	46.8869	1275.971	1306.705
24	81	0	27.1498	29	2	20.2895	1390.063	1417.670
26	80	42	34.6664	29	11	30.0630	2658.768	2684.712
27	81	13	3.0203	29	14	9.8038	572.635	596.770
29	81	20	12.0197	28	59	30.4788	2225.058	2252.632
31	81	13	50.5740	28	36	39.7247	153.388	190.148
32	81	28	37.5055	28	32	25.7803	1550.539	1586.124
33	81	48	59.8159	28	43	11.4156	2763.493	2790.963
34	82	3	45.0593	28	41	54.5121	2151.125	2177.940
35	82	15	42.8377	28	40	40.1534	1848.373	1870.828
36	80	37	44.6546	29	0	56.1884	1308.913	1340.296
37	82	3	16.5838	28	20	22.2409	2375.257	2407.815
38	82	23	37.8634	27	44	33.8903	233.606	275.978
39	82	6	37.6485	28	12	8.7760	628.670	664.365
40	81	53	40.7348	28	3	20.4214	124.567	164.766
41	81	36	57.0578	28	20	13.7834	985.493	1023.412
42	81	43	7.1407	27	59	1.6869	91.831	134.552
43	81	55	21.7043	27	52	3.9930	89.654	132.465
44	81	55	20.1883	27	56	44.0891	408.588	449.559
45	82	15	12.1204	27	51	20.8842	821.926	863.086
46	82	31	49.7716	27	41	27.8349	189.860	232.953
47	82	48	19.0316	27	45	18.4805	938.823	981.000
48	82	47	23.0550	27	50	48.9945	274.898	314.283
49	82	31	32.0864	27	52	17.2396	218.034	257.352
50	82	34	30.0924	28	3	44.6146	1672.743	1707.492
53	82	17	40.3637	28	6	42.6543	598.478	634.027
54	82	57	23.3722	28	21	35.0976	3604.071	3626.878
55	83	9	8.0738	28	18	52.5779	1714.513	1740.309
56	83	1	2.8476	28	5	11.5072	1985.568	2018.415
57	83	8	18.7425	27	55	8.0307	1816.405	1852.530
59	83	23	25.0612	28	14	52.4169	3138.994	3163.246
60	83	26	31.4168	27	56	25.0182	588.154	621.599
61	83	39	37.4888	27	55	17.6658	515.838	549.683
62	83	49	37.7718	27	50	51.5673	1270.104	1304.833
63	83	47	26.5795	28	2	10.6941	881.685	912.704
64	83	48	7.9853	28	9	51.6601	983.404	1011.178
65	83	59	1.9500	28	11	58.6640	791.699	819.083
66	84	5	28.5564	28	1	20.7899	472.948	504.637

67	84	11	14.0112	28	5	1.4130	1252.002	1281.642
68	84	15	26.5422	27	59	45.4486	334.943	367.460
69	84	10	54.7534	27	46	2.4063	1896.329	1933.445
72	84	27	51.1352	28	24	3.2160	3921.863	3939.144
73	84	29	39.2708	28	13	35.4454	1888.355	1911.670
74	84	36	11.0017	28	5	1.3817	1946.339	1972.741
75	85	3	13.6324	28	29	20.1885	3005.323	3016.457
76	80	36	19.6505	29	19	51.6952	1657.162	1682.450
77	82	40	9.6873	28	16	13.6894	2058.608	2088.777
78	81	34	35.5246	28	8	2.2849	107.890	149.882
79	82	37	12.0918	28	27	16.9038	2518.765	2545.625
80	80	23	0.8842	29	44	16.0168	1101.504	1119.674
81	80	32	34.5100	29	50	38.5115	873.339	886.297
82	80	44	1.3495	29	58	41.6117	2494.559	2498.211
83	80	58	55.7568	30	8	2.7678	3630.251	3625.761
84	80	52	23.1370	29	47	40.2444	2448.405	2457.501
85	80	55	30.5077	29	37	41.4576	2553.242	2568.457
86	81	1	30.8885	29	29	31.5762	1871.223	1889.822
87	81	12	58.1134	29	32	26.2539	1790.717	1807.493
88	81	17	15.9593	29	45	11.4520	2525.495	2533.899
89	81	31	20.1559	30	4	52.6195	3175.439	3176.242
91	82	2	40.8488	30	6	41.3822	3899.411	3908.908
92	81	49	15.1967	29	58	15.4743	2962.089	2970.863
93	81	23	18.8916	29	42	6.5115	2799.651	2810.529
94	81	20	57.8315	29	37	5.2206	2626.225	2640.530
95	81	40	12.5628	29	30	11.9751	1386.267	1404.447
96	81	25	51.8255	29	29	44.1385	2246.892	2264.912
97	81	27	28.0669	29	12	28.7091	2450.917	2474.542
98	81	43	13.6681	29	8	28.0747	3048.567	3072.230
99	82	2	3.1183	28	53	54.6985	2275.345	2299.060
100	82	5	14.1243	29	4	54.6802	3027.046	3048.158
101	81	55	26.9649	29	12	57.7174	2541.241	2562.174
102	82	11	30.5623	29	16	25.5378	2358.176	2375.019
103	82	4	55.5796	29	32	21.0406	2961.997	2979.053
104	81	57	13.9166	29	38	16.4738	2623.715	2640.300
105	82	18	21.7561	29	54	43.3323	3531.669	3545.965
106	82	15	35.1250	29	35	5.6381	2768.353	2783.973
107	82	27	21.9911	29	34	53.8390	2950.779	2964.877
108	82	22	11.8917	29	17	21.2606	3117.754	3132.690
109	82	35	27.8934	29	7	16.1088	3018.479	3034.768
110	82	33	21.6927	29	1	29.6993	2133.611	2151.741
112	82	27	4.9395	28	38	13.3303	1554.975	1579.813
113	82	55	32.0838	28	31	2.3998	2812.509	2834.821
114	82	53	4.2152	28	40	19.5257	2412.009	2431.327
115	82	49	16.3974	28	59	7.8224	2486.548	2502.884
117	82	53	55.9597	29	26	33.9395	3933.397	3946.510
118	83	10	54.8265	29	27	0.4638	3887.175	3900.914
119	82	59	42.6398	28	52	21.4662	2816.699	2832.340
120	83	18	47.7687	28	52	19.0621	3952.458	3965.130
121	83	20	12.8473	28	29	30.1295	2017.508	2037.334
122	83	41	5.5337	28	25	8.9204	2536.851	2557.462
123	83	38	5.9377	28	38	12.4519	2447.280	2461.698
124	83	43	28.6561	28	46	50.6430	2723.621	2736.401
125	83	51	45.6997	29	0	7.5001	3471.242	3484.175
126	84	0	51.5880	29	2	30.1129	3923.632	3936.773
127	83	57	42.0193	29	10	56.4801	3795.031	3809.443
128	84	5	28.9803	28	38	28.5049	3370.314	3381.121
129	84	20	28.3258	28	32	8.8768	2149.732	2162.518
130	84	28	22.5002	28	38	5.5074	3693.244	3703.323
131	84	38	26.7986	28	35	28.8590	3547.570	3558.746
133	84	50	2.7107	28	31	28.6418	2391.756	2403.915
134	85	6	28.8557	28	36	46.4741	3810.687	3819.804
136	83	11	24.3777	29	7	48.7458	4084.033	4097.348
137	83	17	15.4352	29	22	14.6495	4159.886	4173.634
138	82	56	25.8893	29	10	21.8509	3617.642	3631.283
139	84	7	7.8479	28	20	7.6274	1756.364	1776.695
333	81	47	43.0255	28	43	55.1613	2616.267	2645.138